

Vancomycert: A Certified Neuro-Symbolic Drug Delivery System

UK Robotics Summer School 2026

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University of Southampton

Overview



- ▶ Problem description
- ▶ Mathematical Formalisation
- ▶ Vehicle Specification
- ▶ Summary



Outline

Problem Description

Mathematical Formalisation

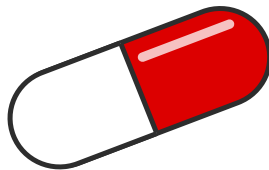
Vehicle Specification

Summary

Vancomycin



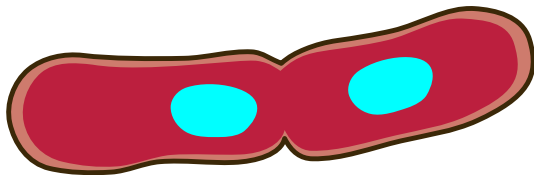
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Vancomycin

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- ▶ Treats severe and resistant bacterial infections



Vancomycin



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- ▶ Treats severe and resistant bacterial infections
- ▶ widely perscribed



Vancomycin

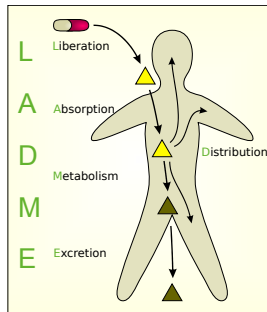
- ▶ Vancomycin is an antibiotic
- ▶ Treats severe and resistant bacterial infections
- ▶ widely perscribed
- ▶ 31.7% of patients experienced nephrotoxicity¹

¹Al-Maqbali et al. – “Vancomycin Therapeutic Drug Monitoring (TDM) and Its Association with Clinical Outcomes”



Pharmacokinetics and Pharmacodynamics Modelling

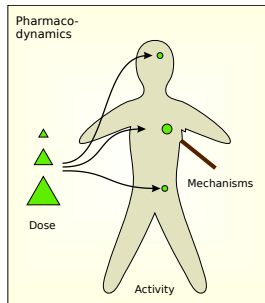
- Pharmacokinetics is what the body does to the drug





Pharmacokinetics and Pharmacodynamics Modelling

- ▶ Pharmacokinetics is what the body does to the drug
- ▶ Pharmacodynamics is what the drug does to the body



Pharmacokinetics and Pharmacodynamics Modelling



► Pharmacodynamics

Pharmacokinetics and Pharmacodynamics Modelling



► Pharmacodynamics → Optimisation

Pharmacokinetics and Pharmacodynamics Modelling



► Pharmacodynamics → Optimisation → Machine Learning



Pharmacokinetics and Pharmacodynamics Modelling

- ▶ Pharmacodynamics → Optimisation → Machine Learning
- ▶ Pharmacokinetics



Pharmacokinetics and Pharmacodynamics Modelling

- ▶ Pharmacodynamics → Optimisation → Machine Learning
- ▶ Pharmacokinetics → Safety-Critical



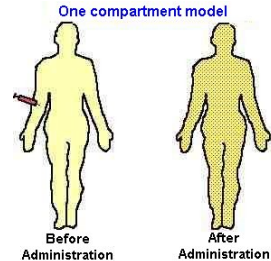
Pharmacokinetics and Pharmacodynamics Modelling

- ▶ Pharmacodynamics → Optimisation → Machine Learning
- ▶ Pharmacokinetics → Safety-Critical → [Formal Methods](#)



Compartment Models

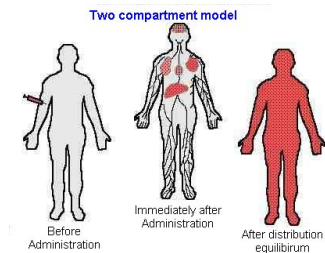
- **One Compartment:** The body acts as one single system, the drug is absorbed by the entire body.





Compartment Models

- ▶ **One Compartment:** The body acts as one single system, the drug is absorbed by the entire body.
- ▶ **Two Compartment:** The organs first absorb the drug and deposit it in the blood, then the rest of the body processes it.





Compartment Models

- ▶ **One Compartment:** The body acts as one single system, the drug is absorbed by the entire body.
- ▶ **Two Compartment:** The organs first absorb the drug and deposit it in the blood, then the rest of the body processes it.
- ▶ We focus on the One Compartment model for simplicity



References

-  Al-Maqbali, Juhaina Salim et al. “Vancomycin Therapeutic Drug Monitoring (TDM) and Its Association with Clinical Outcomes: A Retrospective Cohort”. In: Journal of Infection and Public Health 15.5 (May 1, 2022), pp. 589–593. ISSN: 1876-0341. DOI: 10.1016/j.jiph.2022.04.007. URL: <https://www.sciencedirect.com/science/article/pii/S1876034122000910> (visited on 06/17/2026).



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Pharmacokinetic Constants

Constants	Representation
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Pharmacokinetic Constants

Constants	Representation
k_a	Absorption Coefficient



Pharmacokinetic Constants

Constants	Representation
k_a	Absorption Coefficient
k_e	Elimination Coefficient



Pharmacokinetic Constants

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k_a	Absorption Coefficient
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V_d	Volume Distribution



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Variables	Representation
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Pharmacokinetic Constants

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Variables	Representation
C	Concentration of Vancomycin



Pharmacokinetic Constants

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k_a	Absorption Coefficient
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Vd	Volume Distribution
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Variables	Representation
C	Concentration of Vancomycin
C_Σ	Total Concentration



Pharmacokinetic Constants

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k_a	Absorption Coefficient
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Pharmacokinetic Constants

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k_a	Absorption Coefficient
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D	Dose size
$\bar{D}$	List of all doses



Pharmacokinetic Constants

Constants	Representation
k_a	Absorption Coefficient
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Vd	Volume Distribution
ttd	Time between doses

Variables	Representation
C	Concentration of Vancomycin
C_Σ	Total Concentration
D	Dose size
$\bar{D}$	List of all doses
t	Time size first dose



One Compartment Equation

$$C(D, t) = \frac{D \cdot k_a}{Vd \cdot (k_a - k_e)} \cdot \left(e^{-k_e \cdot t} - e^{-k_a \cdot t} \right)$$

One-Compartment Pharmacokinetic model for oral medication²

²Talevi and Bellera – “One-Compartment Pharmacokinetic Model”



One Compartment Equation

$$C(D, t) = \frac{D \cdot k_a}{Vd \cdot (k_a - k_e)} \cdot \left(e^{-k_e \cdot t} - e^{-k_a \cdot t} \right)$$

One-Compartment Pharmacokinetic model for oral medication²

$$C_{\Sigma}(\bar{D}, t) = \sum_{i=0}^n (\max(0, C(\bar{D}_i, t - ttd \cdot i)))$$

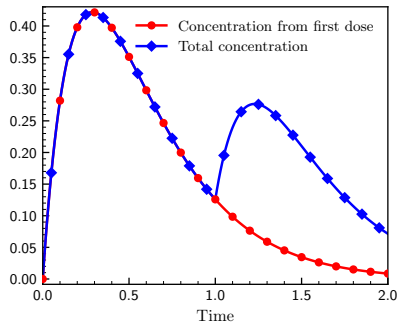
Total concentration³

²Talevi and Bellera – “One-Compartment Pharmacokinetic Model”

³Sirman et al. – Vancomycert: A Certified Neuro-Symbolic Drug Delivery System (Case Study)



One Compartment Equation





Neural Network Involvement

$$\mathcal{N} : \mathbb{R}^5 \rightarrow \mathbb{R}$$



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$\mathcal{N}(C, \text{Temperature}, \text{White blood cell count}, \text{Age}, \text{Weight})$



Neural Network Involvement

$$\mathcal{N} : \mathbb{R}^5 \rightarrow \mathbb{R}$$

$\mathcal{N}(C, \text{Temperature, White blood cell count, Age, Weight})$

C does not depend on any of these⁴

⁴ C does depend on weight, but it is assumed to be constant and known statically



Neural Network Involvement

An **initial dose** is given, then n more doses with interval ttd

$doses: \mathbb{R} \rightarrow \mathbb{N} \rightarrow \text{list } \mathbb{R}$

$$doses(c, n) = \begin{cases} [\mathcal{N}(c)] & n = 0 \\ doses(c, n - 1) ++ [:: \mathcal{N}(C_{\Sigma}(doses(c, n - 1), ttd \cdot n))] & n > 0 \end{cases}$$



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Total Concentration at n^{th} dose



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n^{th} dose



Main Safety Definition

Given that the **concentration** is initially safe, then, for all time, for any number of doses, the total concentration will never exceed the safe boundry.

$$\forall c : \mathbb{R}, c \leq C_{safe} \implies \forall t : \mathbb{R}, \forall n : \mathbb{N}, C_{\Sigma}(doses(c, n), t) \leq C_{safe}$$



Main Safety Definition

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Non-Linear!

Cannot naively be used in vehicle



Property for Vehicle

The **current concentration** is used as an input to the neural network, along with the time at which concentration peaks, can be used to calculate the maximum change (d), which can be added to the current concentration.

$$\forall c : \mathbb{R}, c \leq C_{safe} \implies c + C \left(\mathcal{N}(c), \frac{\ln \left(\frac{k_a}{k_e} \right)}{k_a - k_e} \right) \leq C_{safe}$$



Property for Vehicle

The current concentration is used as an input to the [neural network](#), along with the time at which concentration peaks, can be used to calculate the maximum change (d), which can be added to the current concentration.

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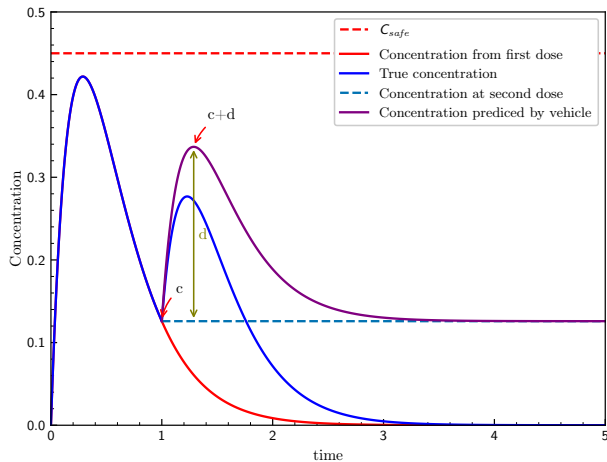
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Property for Vehicle



Not Quite...



$$\forall c : \mathbb{R}, c \leq C_{safe} \implies c + C \left(\mathcal{N}(c), \frac{\ln \left(\frac{k_a}{k_e} \right)}{k_a - k_e} \right) \leq C_{safe}$$



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Solvers can't handle irrational numbers...



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Approximate!



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Solvers can't handle irrational numbers...

Approximate!

$$K_{e_under} \leq e^{-k_e \cdot \frac{\ln\left(\frac{k_a}{k_e}\right)}{k_a - k_e}} \leq K_{e_over}$$

$$K_{a_under} \leq e^{-k_a \cdot \frac{\ln\left(\frac{k_a}{k_e}\right)}{k_a - k_e}} \leq K_{a_over}$$



Not Quite...

If $k_a < k_e$:

$$e^{-k_e \cdot \frac{\ln\left(\frac{k_a}{k_e}\right)}{k_a - k_e}} - e^{-k_a \cdot \frac{\ln\left(\frac{k_a}{k_e}\right)}{k_a - k_e}} \geq K_{e_under} - K_{a_over}$$

else:

$$e^{-k_e \cdot \frac{\ln\left(\frac{k_a}{k_e}\right)}{k_a - k_e}} - e^{-k_a \cdot \frac{\ln\left(\frac{k_a}{k_e}\right)}{k_a - k_e}} \leq K_{e_over} - K_{a_under}$$

The change in sign is to respect the effect of the fraction in C .

$$C(D, t) = \frac{D \cdot k_a}{Vd \cdot (k_a - k_e)} \cdot \left(e^{-k_e \cdot t} - e^{-k_a \cdot t} \right)$$



References



Sirman, Alistair et al.

Vancomycert: A Certified Neuro-Symbolic Drug Delivery System (Case Study).

2026. arXiv: 2606.19532 [cs.LO]. URL: <https://arxiv.org/abs/2606.19532>.



Talevi, Alan and Carolina L. Bellera. “One-Compartment Pharmacokinetic Model”. In:

The ADME Encyclopedia: A Comprehensive Guide on Biopharmacy and Pharmacokinetics

Cham: Springer International Publishing, 2021, pp. 1–8. ISBN: 978-3-030-51519-5.

DOI: 10.1007/978-3-030-51519-5_58-1. URL:

https://doi.org/10.1007/978-3-030-51519-5_58-1.



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Mathematical Formalisation

Vehicle Specification

Summary



Preamble

```
type UnnormalisedInputVector = Tensor Real [5]
```

```
type InputVector = Tensor Real [5]
```

```
conc = 0
```

```
temp = 1
```

```
wbc = 2
```

```
age = 3
```

```
weight = 4
```

```
type OutputVector = Tensor Real [1]
```



Normalisation

```
@dataset
```

```
meanScalingValues : UnnormalisedInputVector
```

```
@dataset
```

```
standardDeviationValues : UnnormalisedInputVector
```

```
normalise : UnnormalisedInputVector -> InputVector
```

```
normalise x = foreach i .
```

```
  (x ! i - meanScalingValues ! i) / (standardDeviationValues ! i)
```



Network

```
@network
```

```
pk : InputVector -> OutputVector
```

```
normpk : UnnormalisedInputVector -> OutputVector
```

```
normpk x = pk (normalise x)
```



Parameters

```
@parameter
Ka, Ke, Vd, C_safe, ttd,
  Ka_over, Ka_under,
  Ke_over, Ka_under : Real4
```

```
@property
Ka_pos : Bool
Ka_pos = 0 < Ka
```

```
@property
Ke_pos : Bool
Ke_pos = 0 < Ke
```

```
@property
Ke_n_Ka : Bool
Ke_n_Ka = Ka != Ke
```

```
@property
Vd_pos : Bool
Vd_pos = 0 < Vd
```

```
@property
C_safe_pos : Bool
C_safe_pos = 0 < C_safe
```

⁴This is not valid syntax, but it is shortened for ease of reading.



Precondition

```
safeInput : Input -> Bool
safeInput x =
  0      <= x ! conc      <= C_safe and
  36.5   <= x ! temp      <= 40    and
  7.5    <= x ! wbc       <= 20    and
  18     <= x ! age       <= 89    and
  50     <= x ! weight    <= 100
```



Main Safety Property

```
safeOutput : Input -> Bool
safeOutput x =
  let y = (((normpk x) ! 0) * Ka) / (Vd * (Ka - Ke)))
  in if Ka < Ke
    then (x ! conc) + y * (Ke_under - Ka_over) <= C_safe
    else (x ! conc) + y * (Ke_over - Ka_under) <= C_safe

@property
safe : Bool
safe = forall x . safeInput x => safeOutput x
```



Non Negativity

```
nonNegOutput : Input -> Bool  
nonNegOutput x = 0 <= (normpk x) ! 0
```

```
@property  
nonNeg : Bool  
nonNeg = forall x . safeInput x => nonNegOutput x
```

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Vehicle for more than just robustness



- ▶ Parametric over patients

Vehicle for more than just robustness



- ▶ Parametric over patients
- ▶ Normalisation

Vehicle for more than just robustness



- ▶ Parametric over patients
- ▶ Normalisation
- ▶ Integration with proof assistants

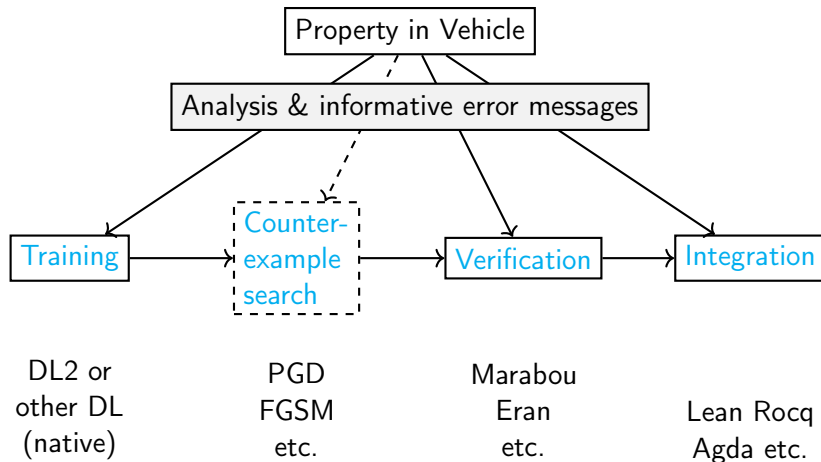


Vehicle for more than just robustness

- ▶ Parametric over patients
- ▶ Normalisation
- ▶ Integration with proof assistants
- ▶ **Property Driven Training**



Vehicle for more than just robustness



Why do all this?



► Ease of use

Why do all this?



- ▶ Ease of use
- ▶ Formalises requirements

Why do all this?



- ▶ Ease of use
- ▶ Formalises requirements
- ▶ Standardisation

Why do all this?



- ▶ Ease of use
- ▶ Formalises requirements
- ▶ Standardisation
- ▶ **Mathematical proofs**

Vancomycert vs AcasXu



- ▶ 1-step vs Infinite-horizon property

Vancomycert vs AcasXu



- ▶ 1-step vs Infinite-horizon property
- ▶ Many networks working in tandem vs one at a time

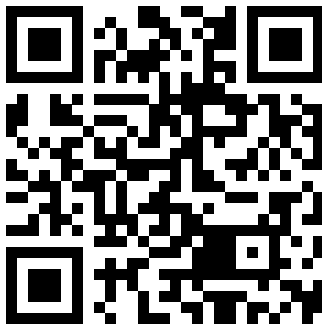
Vancomycert vs AcasXu



- ▶ 1-step vs Infinite-horizon property
- ▶ Many networks working in tandem vs one at a time
- ▶ Both safety critical



Find the paper



Sirman et al. – Vancomycert: A Certified Neuro-Symbolic Drug Delivery System (Case Study)

Next Up...



How to train a network that meets the specification